Sreevaatsav Bavana

Hyderabad, Telangana, India · *Portfolio:* ***sreevaatsavb.github.io/portfolio***

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Personal Profile

Gold medalist in B.Tech in Artificial Intelligence at Mahindra University; currently a Senior Data Scientist at Penguin AI. Actively learning and working on robotic learning with a background in multimodal AI. Research interests include long-horizon task planning for contact-rich manipulation and reinforcement learning (RL) post-training of robot policies.

Education

Mahindra University, Hyderabad, India *Aug 2020 - Aug 2024*

Bachelor of Technology in Artificial Intelligence

- **Gold Medalist** of the 2020–2024 B.Tech AI batch
- **Cumulative GPA** : 8.76
- **Relevant Courses:** Introduction to Robotics, Control Theory, Reinforcement Learning and Autonomous Systems, Signals & Systems, Neural Networks and Deep Learning, Machine Learning and Python, Optimization Techniques for AI, Data Structures

Publications

GIV-CXR: Densely Grounded, Visually Interpretable, Chest X-Ray Question Answering Dataset *2026*

Sreevaatsav Bavana, Adit Rushil Potta, Sai Amrit Patnaik, Nidhi Goyal

MedReasoner Workshop at CVPR

From Layoff to Reemployment: Analyzing Social Media Narratives *2025*

Yayati Gupta, Sreevaatsav Bavana, Sudhira Chegu, Sujith Sai Kalakonda, Yash Mathur, Manmitha Polsani, Sanatan Sukhija

Computational Data and Social Networks (CSoNet 2025)

A Systematic Study on Video Summarization: Approaches, Challenges, and Future Directions *2023*

Kajal Kansal, Nikita Kansal, Sreevaatsav Bavana, Bodla Krishna Vamshi, Nidhi Goyal

Proceedings of the 2nd Workshop on User-centric Narrative Summarization of Long Videos

Projects

Imitation-Learning Policies on a Fairino FR5 Cobot *Mar 2026 - Present*

Sliact Robotics (stealth startup), Founding Engineer · **Write-up: 3-part blog series**

- Collected 1,021 teleoperated demonstrations (∼11.5 hours) on a 6-DOF Fairino FR5 for colour sorting, sanding and pick-to-light, first with an SO-101 leader arm and later a Meta Quest controller over ROS 2. Synced the 100 Hz robot logs with 30 Hz frames from a wrist-mounted RealSense D405 and a scene camera.
- Trained ACT, Diffusion Policy, Octo, π_0 and $\pi_{0.5}$ and ran them closed-loop on the arm at 30 Hz. Fully fine-tuned π_0 on an A100, starting the sanding run from the colour-sorting checkpoint.
- π_0 succeeded in 80% of sanding and 65% of colour-sorting trials (20 autonomous runs each). XR-1 adaptation is in progress.
- Connected π_0 to a VLM planner (Sonnet 5.5) for multi-step tasks and found that neither could identify contact issues. Now adding a force/torque sensor and trying RL post-training with human corrections for sanding and screwing.

AnimalCLEF 2026: Wildlife Re-identification (9th Place) *Mar 2026 - May 2026*

AnimalCLEF 2026 Challenge

- Grouped ∼2,400 unlabeled images by individual animal across 4 species (lynx, salamanders, sea turtles, Texas horned lizards).
- Combined global embeddings from MiewID v3 and MegaDescriptor with local feature matching (KAZE, ALIKED + LightGlue).
- Fine-tuned MegaDescriptor with ArcFace loss (+6.4%) and used SAM2 to mask out background keypoints before matching.

GIV-CXR: Densely Grounded, Visually Interpretable, Chest X-Ray Question Answering Dataset *Aug 2024 - Jan 2025*

VLaNC Lab, Mahindra University (Dr. Nidhi Goyal) · in collaboration with IIIT Hyderabad

- Built a chest X-ray question-answering benchmark with 355,293 question-answer pairs, each tied to bounding boxes across 36 anatomical regions, to test spatial reasoning.
- Set up a hallucination filter that combines automatic checks with radiologist review, with an 82.4% acceptance rate.
- Trained grounded VLM baselines. The best one reached 68.1 mIoU on spatial grounding, nearly double GPT-4o-mini's 39.0, and the top G-Eval answer score (3.98).

Query-Based Video Summarization with a Learnable Text-to-Frame Projection *Aug 2023 - Dec 2023*

Supervisor: Dr. Nidhi Goyal, Assistant Professor, Mahindra University

- Added a learnable GELU projection that maps CLIP text-query embeddings into the video-frame embedding space of Moment-DETR, a transformer that finds the moments and highlights in a video that match a text query.
- Trained it on a subset of QVHighlights and improved mAP by ∼18% over the original Moment-DETR.

Work Experience

Senior Data Scientist *May 2026 - Present*

Penguin AI, Hyderabad, India

- Lead the AI module of a prior-authorization agent system for a large US health plan, built with LangGraph and AWS Bedrock. It has processed 18,500+ cases across 110 clinical guidelines with 99.4% pipeline success and a 4.4-minute median latency.
- Built a guideline selector based on self-consistency sampling that corrected 830 of 1,185 audited routing decisions. With this and other accuracy fixes, agreement between the AI and human reviewers rose from 31% to 67% per procedure line in four months (Orthopedics: 47% to 77%), and unsafe approvals stayed under 1%.
- Helped raise turnaround-time compliance from 44% to 92% and cut reviewer time per line by nearly 40% (4.4 to 2.8 min) across 41 clinical reviewers.

Data Scientist *Jul 2025 - May 2026*

Penguin AI, Hyderabad, India

- Built a claim-denial prediction system end to end, from per-client ETL to a LightGBM and XGBoost ensemble trained with MLflow, with PSI drift alerts and periodic retraining. It reached 0.87 AUC and caught 92% of denials on 18,823 held-out claims.
- Built an LLM agentic workflow that turns payer medical-policy PDFs into questionnaire trees. It scored higher question F1 than the existing drafts (0.64 vs 0.60), at $0.30 to $0.60 and 5 to 10 minutes per policy.

Associate Data Scientist *Jul 2024 - Jun 2025*

Awone AI, Hyderabad, India

- Made production LLM inference 15% faster with quantization and an n-gram speculative decoding head.
- Built a medical PII redaction system that combined a BERT NER model, trained with a custom loss, with document vision-language models. It redacted correctly 90% of the time, against 60-70% for standard tools.
- Trained document vision-language models to fill in forms automatically, with 82% accuracy on forms they had not seen before.
- Built a voice chat app from open-source components that users can interrupt mid-reply, plus an MCP client and server for querying a user database.

Data Science Intern *Jan 2024 - Jun 2024*

Awone AI, Hyderabad, India

- Re-implemented Llama-2, Mistral, Qwen2 and LLaVA from scratch to understand their architectures and attention.
- Explored generative and multimodal models, including diffusion models built from scratch, Stable Diffusion 3, speech LLMs that output audio, and PaliGemma.
- Studied ways to speed up LLM inference, such as speculative decoding with n-gram and Medusa heads, KV caching, FlashAttention and quantization.

Achievements

9th place in AnimalCLEF 2026 Wildlife Re-identification Challenge *2026*

Merit scholarship for all four years (2020-2024) for ranking in the top 5% of the branch *2024*

3rd place in ICETCI Hackathon 2023, Mahindra University *2023*

Rank 111 of 4,000+ teams in Amazon ML Challenge 2023 *2023*

Courses & Certifications

Modern Robotics, Course 1: Foundations of Robot Motion *Apr 2026*

Coursera, Northwestern University · Grade: 98.46%

Modern Robotics, Course 2: Robot Kinematics *In Progress*

Coursera, Northwestern University

Skills

Languages & Libraries	Python, SQL, Pandas, NumPy, Scikit-learn, HuggingFace, PyTorch, OpenCV, C++
AI Engineering & Infrastructure	LangChain, LangGraph, vLLM, MLFlow, Ray
Cloud & Deployment	AWS, Docker, Async programming, Linux, GitHub Actions, MongoDB, PostgreSQL, Kubernetes
Robotics & Simulation	ROS 2, Robot kinematics, MuJoCo, CoppeliaSim